Dice Dreams: Product Analytics & Behavior Deep-Dive

An End-to-End Data Project: From Event Taxonomy to Actionable Insights

Project Overview

This comprehensive project simulates the role of a Gaming Product Analyst. It explores the full data lifecycle of the social-casual title Dice Dreams, focusing on optimizing the player journey, deconstructing the core game loop, and identifying key levers for retention and monetization.

Core Focus Areas

- **Tracking Design:** Developing a robust Event Taxonomy and Data Schema.
- **Onboarding Optimization:** Funnel analysis of the First Time User Experience (FTUE).
- **Behavioral Analytics:** Analyzing the "Build & Raid" core loop through synthetic datasets.
- **Business Intelligence:** Visualizing KPIs to drive actionable product recommendations.

Technical Documentation

- **Full Event Dictionary:** Google Sheets | *A detailed breakdown of triggers and attributes.*
- **Interactive Dashboard:** Looker | *Live visualization of player trends.*
- **SQL Repository:** GitHub | *Optimized queries for the BigQuery environment.*

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Tools Used: SQL (BigQuery), Looker Studio, Python, Google Sheets.
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Project Abstract & Product Overview

Selection Rationale & Business Context

As a passionate gamer aspiring to become an analyst in the gaming industry, I chose to analyze Dice Dreams due to its fast paced onboarding and high social engagement. My objective is to map and optimize User Journey, focusing on how players transition from initial install to long term active users.

This project focuses on Product Analytics rather than economy. By analyzing core growth metrics such as DAU,Installs and Retention, I aim to identify friction points within the user experience. By leveraging data driven insights, I aim to provide actionable recommendations for improving user engagement and funnel conversion.

Product Definition & Core Purpose

Dice Dreams is a mobile **"Social-Casual"** title centered around a Build & Raid loop. Players roll dice to earn coins, which are then invested in building and upgrading themed kingdoms. The core purpose is to provide a high engagement entertainment experience through social interactions, such as attacking a friends' kingdom or stealing their resources, creating a strong psychological incentive for daily login.

Business Model & Revenue Strategy

The application follows a **Freemium** model. While the core game is free to play **(F2P)**, revenue is generated primarily by **In App Purchases (IAP).** The monetization strategy targets "bottlenecks" in the gameplay loop, selling extra Dice Rolls and resource bundles for players who want to accelerate their progression, complete buildings faster or retaliate against competitors.

Target Audience & Market Positioning

The primary audience for Dice Dreams consists of "social-Casual" gamers, characterized by frequent, short burst gameplay sessions and a strong preference for social competition. I discovered the application through its high visibility in the Google Play Store charts, where it consistently ranks as a **Top 9 Grossing** game in the Casual category and remains among the **Top 50** titles across all gaming genres globally. This elite market positioning reflects its immense commercial success and the effectiveness of its long-term player retention strategies.

Onboarding Funnel & Conversion Analysis

The Strategic Role of Onboarding in Dice Dreams

In the “Social-Casual” market, the onboarding process is a primary filter for new player quality. While long term retention is driven by the “Build & Raid” loop and social mechanics, the tutorial serves as the initial conversion point within the user journey. An efficient onboarding experience is a vital indicator of a player’s potential to transition into a long term active user. It ensures that players successfully reach the “Aha! Moment” where they first experience the core game loop. For product analytics and user acquisition (UA) purposes, tutorial completion is a high value early signal allowing us to identify high intent cohorts and detect immediate friction points before they escalate into churn.

Core Tutorial Objectives & Measurement

To ensure the effectiveness of the player’s first experience, the analysis focuses on three key performance pillars:

- **Conversion Efficiency:** Tracking the percentage of players who successfully navigate the guided path to become active users.
- **Friction & Barriers:** Identifying specific mechanical or technical hurdles that disrupt the gameplayflow
- **Time to Value:** Monitoring the duration of the onboarding process to ensure players reach the core gameplay experience the full “Roll,Build,Raid” cycle as quickly as possible.

Synthetic Dataset Structure & Data Schema

The quantitative phase of this project utilizes a synthetic dataset designed to mirror a production-grade tracking plan. The data captures granular behavioral signals through a structured schema:

- **Identity & Context:** Includes unique identifiers for users, sessions, and individual events, alongside application versions to allow for cross-version performance benchmarking.
- **Temporal Markers:** Each interaction is timestamped, enabling the calculation of time spent on individual steps and the total duration of the onboarding session.
- **Funnel & Progression:** The dataset tracks the journey through 9 defined stages, ranging from initial entry and profile creation to feature discovery, social account integration, and final completion.
- **Friction & Navigation:** Technical and user-driven errors are recorded by type, including the time required for the user to recover. It also captures the navigation path between steps and the specific point of exit for non-completers.
- **Outcome Indicators:** A logical flag confirms whether the process was successfully finished, serving as the primary metric for funnel conversion.

The Rationale for Tutorial Event Logging

In mobile gaming, the tutorial is the first high-friction environment a player encounters. To effectively optimize this experience, it is critical to implement a robust event logging strategy. Detailed logging allows analysts to transform a linear sequence of screens into a measurable funnel. By capturing specific behavioral signals at every milestone, we can pinpoint exactly where users struggle, identify technical bottlenecks, and measure the "velocity" of a player's journey toward the core game loop. Without granular event data, it is impossible to distinguish between a player who left due to a lack of interest and one who left due to a specific UI friction point.

Core Event Definition: tutorial_step_completed

To map the onboarding journey, I have defined the tutorial_step_completed event as the primary metric for progression. This event is triggered every time a player successfully navigates from one tutorial phase to the next.

The event holds the following key attributes derived from the synthetic dataset:

- **step_name (String)**: The unique descriptive name of the milestone completed (e.g., Arrival_Kingdom_1).
- **step_index (Integer)**: The sequential position of the step in the tutorial flow, starting from 0.
- **time_spent (Float)**: The duration in seconds taken to complete this specific step.
- **total_time (Float)**: The cumulative time in seconds since the very start of the tutorial until this step was finished.

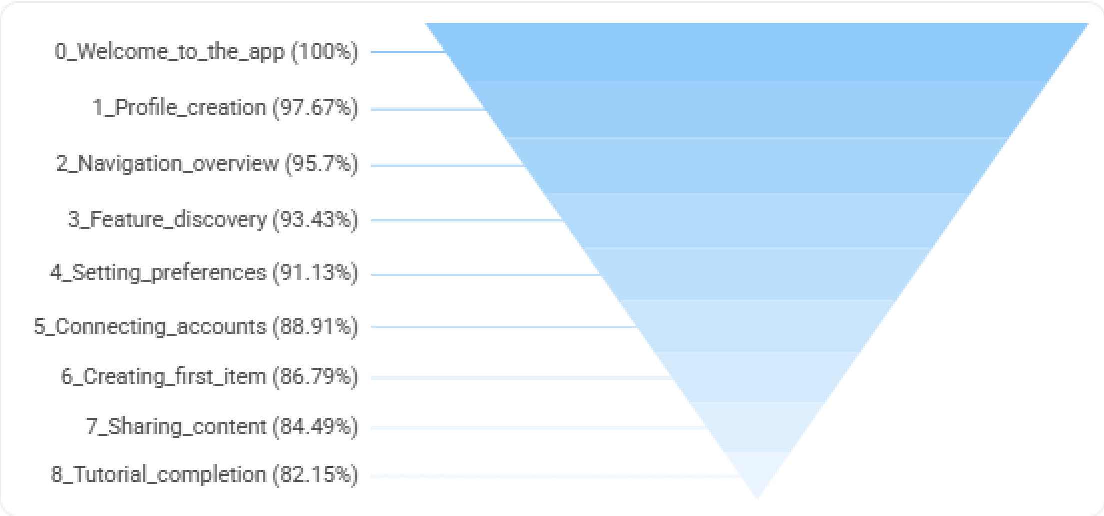
Data Entry Example from the Synthetic Table

Using a specific record from the dataset, we can see how a progression milestone is captured:

- **User ID**: aed6e2f7
- **Event ID**: 42b0c9dc
- **Event Type**: tutorial_step_completed
- **App Version**: 2.0.0
- **Tutorial ID**: main_onboarding
- **Step Index**: 0
- **Time Spent on Step**: 16.426197705797794 seconds

Visualization: The Onboarding Funnel

To monitor the onboarding phase effectively, I utilized a **Funnel Chart** as the primary visualization tool. This chart provides a clear view of user progression through the key milestones, making "leaks" in the funnel immediately visible.



Synthetic Data Analysis & Trends

The analysis of the synthetic data reveals a significant optimization trend across application versions. In earlier iterations, the data indicates high abandonment rates during the initial navigation and configuration stages. These friction points were characterized by elevated error counts and prolonged recovery periods, which directly correlated with lower overall conversion rates.

In the latest versions of the dataset, there is a measurable improvement in process velocity. The reduction in technical friction particularly during the feature discovery and content creation phases has led to a substantial increase in completion rates. The data confirms that as users encounter fewer technical hurdles, the "Time-to-Value" decreases, allowing for a more efficient transition into the core game environment.

To explore these trends and interact with the full dataset including version by version comparisons and error distribution heatmaps you are invited to **view the interactive dashboard**.

Core Gameplay & Analytics Framework

The Role of Event Logging in Game Analysis

Event logging is the foundation for understanding how players actually interact with the game. By capturing specific user actions, we transform raw gameplay into data that allows us to measure engagement, identify friction points, and balance the game economy. Analyzing this data helps us calculate key metrics like conversion rates and retention. Ultimately, a well-defined tracking plan bridges the gap between seeing what happened and understanding why it happened, providing the insights needed to improve the player experience.

Top 10 High-Impact Events

To monitor the overall health of the game, we track events across the entire player journey, focusing on the core "Spin-Earn-Build" loop:

1. **session_start:** Measures Daily Active Users (DAU) and session frequency.
2. **dice_roll_completed:** The core engagement action, essential for balancing the game economy.
3. **attack_initiated:** Tracks the intent to compete and building selection.
4. **attack_finished:** Measures the outcome of attacks, including Bullseye hits and card bonuses.
5. **steal_started:** Records when a player starts the "Push-Your-Luck" mini-game and the potential prize.
6. **steal_finished:** Tracks the final amount stolen and the "Bomb" explosion rate.
7. **raid_finished:** Measures the success rate in the obstacle course and the impact of Raid cards.
8. **card_collected:** Records the acquisition of cards, specifically tracking "New Card" vs. "Duplicate."
9. **kingdom_building_finished:** The primary indicator of player progression and coin sink.
10. **out_of_rolls:** A critical conversion point that triggers shop entries or churn.

Event Definitions & Attributes

Below is a detailed definition of 7 key events, including the specific attributes derived from the Tracking Plan:

I. attack_finished

Triggered after the attack animation ends and the reward is shown.

- **outcome (String):** Result of the attack (e.g., Success, Blocked).
- **is_bullseye (Boolean):** Whether the player hit the center target for a bonus.
- **card_level_bonus (Float):** The multiplier applied based on the player’s Attack Card level.

- **total_gold_gained (Integer):** Final amount after all multipliers and bonuses.
- *Example: outcome: "Blocked", is_bullseye: true, card_level_bonus: 1.5, total_gold_gained: 150000*

II. steal_finished

Triggered when the player stops rolling or hits a bomb in the Steal mini-game.

- **is_exploded (Boolean):** Whether the player hit the bomb and lost part of the prize.
- **total_tries (Integer):** How many times the player rolled the steal-die.
- **final_amount_stolen (Integer):** Final coins added to balance after the card level bonus.
- *Example: is_exploded: false, total_tries: 4, final_amount_stolen: 850000*

III. raid_finished

Triggered when the raid obstacle course ends.

- **total_obstacles_passed (Integer):** How many obstacles were successfully cleared.
- **is_grand_prize_reached (Boolean):** Did the player reach the end of the course.
- **total_raid_coins (Integer):** The sum of all coins earned, including the Raid card bonus.
- *Example: total_obstacles_passed: 3, is_grand_prize_reached: true, total_raid_coins: 1200000*

IV. dice_roll_completed

Triggered once the dice land and the reward is calculated.

- **multiplier (Integer):** The bet multiplier used for the roll (e.g., 1, 3, 5, 10).
- **reward_type (String):** The outcome (e.g., Attack, Steal, Raid, Coins).
- **is_super_roll (Boolean):** Whether a special event multiplier was active.
- *Example: multiplier: 10, reward_type: "Steal", is_super_roll: false*

V. card_collected

Triggered when a player receives a card from any source.

- **card_id (String):** Unique identifier for the specific card.
- **is_new_card (Boolean):** Critical for tracking collection progression vs. duplicates.
- **source (String):** How the card was obtained (e.g., Chest, Event, Gift).
- *Example: card_id: "golden_shield_05", is_new_card: true, source: "Chest"*

VI. out_of_rolls

Triggered when a player's roll balance reaches zero during an interaction.

- **last_multiplier (Integer):** The multiplier used just before running out.

- **offer_presented (String):** The immediate offer shown (e.g., Small_Pack, Ad_Reward).
- **interaction (String):** User reaction (e.g., Clicked, Dismissed).
- *Example: last_multiplier: 5, offer_presented: "Ad_Reward", interaction: "Clicked"*

VII. team_help_rolls_given

Triggered when a user provides rolls to a teammate.

- **teammate_user_id (String):** The ID of the recipient.
- **reward_for_helper (Integer):** The coins/bonus given to the user for helping.
- **team_id (String):** The identifier of the player's squad.
- *Example: teammate_user_id: "dd_8822", reward_for_helper: 5000, team_id: "Israel_Kings"*

Global Event Attributes (Super Properties)

The following attributes are included in every single event for cross-segmentation:

- **user_id:** Unique player identifier.
- **timestamp:** Exact time of the event (UTC).
- **app_version:** Current build version (e.g., 2.1.13).
- **platform:** Operating system (iOS/Android).
- **device_model:** Hardware used (e.g., Redmi Note 13 Pro).
- **country_code:** Geographic location (ISO 2).
- **player_level:** Current Kingdom level.
- **rolls_balance:** Current number of rolls at the time of the event.
- **coins_balance:** Current amount of coins at the time of the event.

Strategic Analytics & Performance Metrics

Data-Driven Performance Monitoring

Establishing and monitoring a robust set of health metrics is essential for evaluating the game's operational success and long-term sustainability. By tracking these indicators, we can differentiate between surface-level growth and true product-market fit. These metrics allow the product and data teams to identify friction points in the player journey, optimize the economic balance, and ensure that the cost of user acquisition is consistently offset by player lifetime value.

To provide a comprehensive view of the game's performance, our KPIs are categorized into three strategic pillars: **Retention & Ecosystem Health** (Staying), **Core Engagement & Gameplay Depth** (Playing), and **Monetization Efficiency** (Paying).

Key Metric Definitions

I. Retention & Ecosystem Health (Staying)

Focuses on the game's ability to maintain a stable and growing player base over time.

- **Retention Rate (D1, D7, D30)** Measures the percentage of players who return to the game after a specific number of days from their install date. It is the primary indicator of long-term engagement and product-market fit.
    - **Formula:**

$$\frac{Number\ of\ unique\ users\ who\ returned\ on\ Day\ X}{Total\ number\ of\ unique\ users\ who\ installed\ the\ app\ on\ Day\ 0} \times 100$$

- **DAU (Daily Active Users)** The total number of unique players who engage with the game in a single 24-hour window. This measures the daily scale and vitality of the player base.
    - **Formula:** The distinct count of unique users who initiated at least one game session within a given day.
- **Stickiness (DAU/MAU Ratio)** Indicates how many of the monthly active players return to the game on a daily basis. High stickiness confirms that the core gameplay loop effectively retains users.
    - **Formula:** $\frac{Daily\ Active\ Users}{Monthly\ Active\ Users}$
- **CPI (Cost Per Install)** The cost incurred to acquire a single new player through marketing efforts. This is essential for determining the profitability of growth campaigns.
    - **Formula:** $\frac{Total\ marketing\ spend}{Total\ number\ of\ new\ installs}$

II. Core Engagement & Gameplay Depth (Playing)

Analyzes how deeply players interact with the mechanics and social features of *Dice Dreams*.

- **Average Session Length** Measures the average duration of a single gameplay experience. This reflects how effectively the game's mechanics keep players occupied.
    - **Formula:**

$$\frac{Total\ cumulative\ duration\ of\ all\ sessions}{Total\ number\ of\ sessions}$$

- **Feature Adoption Rate** Tracks how many players actively engage with specific social or progression features, such as team activities or card collection mechanics.
    - **Formula:**

$$\frac{Number\ of\ unique\ users\ who\ engaged\ with\ the\ feature}{Total\ Daily\ Active\ Users}$$

III. Monetization Efficiency (Paying)

Evaluates the game's ability to convert engagement into sustainable revenue.

- **ARPU (Average Revenue Per User)** Measures the average revenue generated by every user, regardless of whether they have made a purchase or not.
    - **Formula:**

$$\frac{Total\ revenue\ generated\ within\ a\ period}{Total\ number\ of\ unique\ users\ within\ that\ same\ period}$$

- **ARPPU (Average Revenue Per Paying User)** Focuses on the spending behavior of the paying segment, helping to evaluate the profitability of monetization offers.
    - **Formula:**

$$\frac{Total\ revenue\ generated\ within\ a\ period}{Number\ of\ unique\ users\ who\ made\ at\ least\ one\ purchase\ during\ that\ period}$$

- **LTV (Lifetime Value)** The predicted total revenue a single player will generate throughout their entire tenure in the game.
    - **Formula:**

$$Average\ Revenue\ Per\ User\ \times\ Average\ player\ lifespan\ in\ days$$

Data Architecture & Strategic Storage Efficiency

To support scalable and cost-effective strategic analysis, we implement a Summary Table (OLAP Cube) methodology. This approach moves the analytical burden from the raw event stream to a structured, high-performance aggregation layer.

The Method: Multi-Dimensional Aggregation

Instead of querying raw event-level data (Fact Tables), which can contain millions of rows, we utilize an ETL (Extract, Transform, Load) process to materialize data into summary tables. This process translates "Atomic Events" into "Business Metrics" at the required grain (User, Date, or Version).

Advantages & Disadvantages (Pros/Cons)

Feature	Pros (Advantages)	Cons (Disadvantages)
Analysis Speed	Queries on summary tables run in milliseconds, enabling rapid hypothesis testing.	Data is usually processed in batches (e.g., T+1), so it's not live.
Operational Cost	Scanning pre-aggregated data is up to 99% cheaper than scanning raw fact tables.	Requires initial engineering effort to build and maintain the ETL logic.
Logic Integrity	Calculations (like ARPU or Retention) are locked in the code, preventing metric drift.	Changes in business logic require re-calculating historical data (Backfilling).
Data Quality	The process filters out noise and duplicates before the data reaches the analyst.	Loses the extreme granularity of raw logs.

The Pipeline Process: From Raw Events to Strategic Pillars

The data follows a structured three-stage pipeline to bridge the gap between raw logs and business insights:

1. **Ingestion & Flattening**: Raw JSON payloads from the fact table are extracted and flattened. Parameters like app_version, platform, and price_usd are typed and prepared for the transformation layer.
2. **Strategic Pillar Categorization:**
   - **Staying (Retention)**: The pipeline identifies the install_date for every user and calculates the "Day-N" distance for each active day to build a retention matrix.
   - **Playing (Engagement)**: Gameplay-specific events (e.g., sessions, tutorial steps) are aggregated to define the depth of player interaction.
   - **Paying (Monetization)**: Transactional data is normalized to USD and aggregated into daily and cumulative revenue metrics (LTV).
3. **Materialization:** The enriched data is saved into a User-Day Grain table. This serves as the single source of truth for all strategic pillars, eliminating the need for expensive joins during analysis.

Dashboard Architecture & Data Visualization

The Role of the Dashboard in Business Operations

The *Dice Dreams* analytics dashboard serves as the central command center for all stakeholders, consolidating complex data streams into actionable visual insights. By providing a single source of truth, the dashboard enables the team to monitor the game's "pulse" in real-time, identify emerging trends, and evaluate the impact of product changes. The architecture of the dashboard is designed to facilitate a top-down analysis—starting with high-level business health and drilling down into granular user behavior and long-term retention patterns.

Dashboard Layout & Components

The dashboard is meticulously organized into three specialized views, each serving a distinct analytical purpose in the player lifecycle.

Page 1: Gaming Performance Overview

This page provides a macro view of the game's commercial health and user growth, focusing on key performance indicators (KPIs) and market quality.

- **KPI Scorecard:** A top-level summary featuring cumulative performance for **Today's Revenue**, **Total Installs**, **Active Users**, and **Avg. ARPU**.
- **Daily Growth & Engagement Trends:**
    - **Total Installs:** A time-series chart monitoring acquisition volume to identify spikes from marketing campaigns.
    - **Daily Active Users (DAU):** A trend line tracking engagement patterns over time.
- **Market Quality & Revenue Distribution:**
    - **Geographic Acquisition Heatmap:** A global map visualizing revenue concentration and install volume by region.
    - **ARPU by Country:** A bar chart identifying high-value markets (e.g., KR, RU, BR) to guide regional optimization.

Page 2: Product & Monetization Deep-Dive

This view analyzes the game's internal economy and how players progress through "Villages" relative to their spending habits.

- **Village Economy & Scaling:**
    - **User Segment Distribution:** A bar chart breaking down player tiers (Minnow, Dolphin, Whale) across game stages.
    - **LTV vs. Current Village:** A scatter plot visualizing the correlation between a player's Lifetime Value (LTV) and their progression, helping identify "Whales" who advance rapidly.

- **Product Consumption & Stage Distribution:**
    - **Revenue Mix by Product:** A donut chart showing the share of revenue from **Spins**, **Coins**, **Access & VIP**, and **Offers**.
    - **Product Adoption per Game Stage:** A stacked bar chart visualizing which items are consumed at specific villages, pinpointing where players shift their spending from core gameplay (Spins) to utility (Coins).

Page 3: Player Retention Analysis

The final view focuses on product "stickiness" and the long-term loyalty of user cohorts.

- **User Retention Cohort Heatmap:** A specialized matrix tracking cohorts by their install date. It measures the percentage of returning users across critical intervals (Day 2, 7, 14, 28, and 60). The blue-to-grey gradient allows stakeholders to instantly spot the "cliff" where user drop-off occurs and assess the stability of the long-term player base.

User Interactivity & Data Filtering

To facilitate granular analysis, the dashboard is equipped with dynamic controls:

- **Date Range Selector:** Enables analysis of specific periods, such as holiday events or new version releases.
- **Country Filter:** Allows for deep-dives into specific regional performance (available on the Overview page).
- **Global Sync:** Filters applied to the top-level controls propagate through the relevant charts to ensure a consistent "single source of truth."

Strategic Dashboard Analysis & Actionable Insights

This section outlines the critical findings from the Dice Dreams analytical suite, followed by specific strategic recommendations to improve performance, retention, and monetization.

1. User Acquisition & Market Efficiency

- **Observation:** There is a dramatic drop in daily installs, falling from a steady average of 400 (Oct-Jan) to near-zero in the current month.
- **Analysis:** This suggests high competitor marketing activity or a lapse in our User Acquisition (UA) strategy.
- **Actionable Recommendation:** Conduct a competitor benchmarking analysis for UA campaigns. Immediately shift marketing budget to high-revenue/low-install "Quality Pockets" identified in the Map Chart. **Deprioritize the Japanese market** due to consistently low ARPU and prioritize high-conversion regions to maximize ROI.

2. Village 12 Progression & "The Economy Wall"

- **Observation:** A massive spike in "Coin Purchases" occurs at Village 11 & 12, but player progression almost completely halts immediately after.
- **Analysis:** Village 12 acts as a "Hard Gate" where the cost to progress exceeds player patience, leading to either a forced purchase or churn.
- **Actionable Recommendation: Re-balance the economy for Village 12.** We should lower the building upgrade costs or introduce a "Progress Booster" reward at the end of Village 11. This will smooth the difficulty curve and transition players into the more stable "Mid-Game" cohort rather than losing them at the first hurdle.

3. Early Engagement & D2 Retention Crisis

- **Observation:** Recent Day-2 (D2) retention is alarmingly low, with some cohorts dipping as low as 28.6%.
- **Analysis:** Players are not finding enough value or excitement within the first 24-48 hours to justify a second login.
- **Actionable Recommendation: Optimize the "Early Velocity" experience.** Introduce a "High-Reward Login Bonus" for Day 2 (e.g., massive spin bundles or rare cards). Additionally, simplify the progression in Villages 1-3 to ensure players reach the social "Team" features faster, which is a proven driver for long-term stickiness.

4. Churn Reversal & "Resurrected" Players

- **Observation:** The Retention Cohorts show "non-linear" patterns where retention percentages occasionally increase (players returning after being inactive).
- **Analysis:** There is a latent segment of players who are still interested in the game but need a trigger to return.
- **Actionable Recommendation: Scale the Re-engagement Engine.** Identify the specific push notifications or retargeting ads that brought these players back and automate them into a "Win-Back" campaign for all players who reach Village 5 but become inactive for more than 7 days.

5. Daily Activity (DAU) Stabilization

- **Observation:** DAU has been on a consistent downward trend for the past three months.
- **Analysis:** The lack of new installs combined with high early-game churn is hollowing out the active user base.
- **Actionable Recommendation:** Implement **"Live-Ops Daily Missions."** By introducing short-term, daily objectives with incremental rewards, we can create a stronger daily habit for the remaining user base while the UA team works on fixing the top-of-funnel acquisition.

Accessing the Live Dashboard

The strategic metrics and visual analyses described above are consolidated into an interactive, real-time dashboard. This tool allows for dynamic exploration of the data, enabling stakeholders to filter by specific timeframes, regions, and versions to gain deeper insights into the *Dice Dreams* ecosystem.

Access the live dashboard here:

Link to Interactive Dice Dreams Dashboard

Product Strategy & Feature Specification

The Evolution of Gameplay Mechanics

To maintain long-term player interest and drive sustainable growth, a game must evolve beyond its core loop. Continuous innovation is essential for shifting players from "passive luck" to "active strategy." By introducing new meta-layers, we increase the game's complexity for high-level players, providing them with new goals and meaningful ways to spend their resources.

Feature Proposal: "The Royal Architects"

1. Executive Summary

Currently, progression in *Dice Dreams* is linear: players earn coins to build and complete kingdoms. **"The Royal Architects"** introduces a specialized crew of experts that players can unlock and assign to their kingdoms. These architects provide passive bonuses to building costs, shield durability, and raid rewards, adding a layer of strategic decision-making to every spin.

2. The Problem & Solution

- **Problem:** Late-game players often face "progression fatigue," where building costs become extremely high, and the gameplay feels repetitive.
- **Solution:** Architects allow players to specialize their gameplay. If a player wants to progress faster through kingdoms, they hire the *Master Builder*. If they want to protect their progress from attacks, they hire the *Royal Guard*.

3. User Story

"Player Sarah logs in and receives a 'Blueprint' found during a successful attack. She uses the blueprint to recruit 'Leo the Master Builder.' The game informs her: 'Leo reduces all building costs by 15%!' Sarah assigns Leo to her current kingdom. Now, every coin she earns goes further, allowing her to finish the kingdom faster before her shields run out."

4. Core Mechanics

- **Architect Recruitment:** Players collect "Blueprints" from Raids or Chests.
- **Specializations:**
  - **The Master Builder:** Reduces building costs and upgrade times.
  - **The Royal Guard:** Increases the chance that a blocked attack won't consume a shield.
  - **The Treasure Hunter:** Adds a multiplier to coins stolen during Raids.
- **The Energy System (The Sink):** Architects need **"Inspiration"** (consumable) to remain active. Once their inspiration meter hits zero, their bonus disappears until fed with "Inspiration Sparks" found in-game or purchased.

5. Economy & Monetization

- **Primary Sink (Inspiration):** Players can purchase "Inspiration Bundles" to keep their best architects active during high-stakes events.
- **Secondary Sink (Blueprints):** High-tier blueprints are rare, encouraging players to purchase premium Chests.

6. Visuals & UI

- **Main Screen:** The active architect stands next to the Castle in the current kingdom (or next to the "Build" button).
- **Animation States:**
    - **Active:** The architect is seen working, examining blueprints, or cheering.
    - **Action:** When a bonus is triggered (e.g., a cheaper building), the architect performs a "Success" animation with a gold glow.
    - **Exhausted:** The architect sits down or falls asleep. Tapping them opens the "Feed Inspiration" popup.
- **Architect Menu:** A new tab in the collection menu showing all unlocked specialists and their current levels.

7. Predicted KPI Impact

Metric	Prediction	Reasoning
ARPU	📈+12%	Purchase of "Inspiration" and Blueprint Chests.
Session Length	📈+8%	Strategic management of architects and energy levels.
D30 Retention	📈+6%	Long-term progression goals through architect leveling.

AB Testing & Statistical Analysis

Here is an abstract summary for an A/B test on the new "Royal Architects" feature for Dice Dreams.

Experiment Abstract: "Royal Architects" Rollout

1. Objective & Hypothesis

The primary objective of this experiment is to improve D7 Retention and Average Revenue Per User (ARPU) for mid-to-late game players (Kingdom 15+).

- Hypothesis: We believe that introducing the "Architect" mechanic will provide players with a sense of strategy and agency. By allowing players to optimize their building costs and resource management, we will increase daily engagement (re-visiting to manage energy) and drive monetization through "Inspiration" purchases.

2. Methodology (The Split)

We will perform a randomized **50/50** split on all users who reach Kingdom 15 during the 14-day test period.

- **Group A (Control)**: Users continue playing the standard version of the game without the Architect system.
- **Group B (Variant)**: Upon reaching Kingdom 15, these users unlock the "Blueprint" tutorial and receive their first Architect (Leo the Master Builder). They gain access to the Architect UI and the "Inspiration" energy mechanic.

3. Key Performance Indicators (KPIs)

- **Primary Metric**: **D7 Retention** (Target: Increase from 14% $\rightarrow$ 16%).
- **Secondary Metric**: **ARPU** (Target: Increase due to Inspiration Spark purchases).
- **Guardrail Metric**: **Progression Speed** (Ensure that building cost reductions do not exhaust content too quickly, leading to long-term churn).

4. Duration & Sample Size

- **Duration**: 14 Days.
- **Sample Size**: Minimum 12,000 users per group to achieve 95% statistical significance.

AB Test - Statistics & Results

The Results

- **Group A (Control)**: 14.2% Retention Rate.
- **Group B (Variant)**: 15.8% Retention Rate.
- **Difference**: Group B performed +1.6% better in absolute terms (a relative lift of ~11.2%).

Statistical Significance (Z-Test)

- **P-Value**: 0.00006 (P < 0.001%)
- **Z-Score:** 4.02
- **Significance Level** (α): 0.05 (5%)

Conclusion: Statistically Significant

Since the p-value (0.00006) is significantly lower than the significance level (0.05), and the Z-Score (4.02) is **well above** the critical value of 1.96, we **reject the null hypothesis**.

Statistical Significance & Insight

The observed 1.6% increase in D7 retention is statistically significant. There is a less than 0.01% probability that this improvement occurred due to random chance, confirming that "The Royal Architects" feature is a genuine driver of player retention.

Recommendation

1. **Full Rollout**: Proceed with rolling out the "Royal Architects" feature to 100% of the user base.
2. **Monitor Guardrails**: Closely track the "Kingdom Completion" speed over the next 30 days to ensure the economy remains balanced.